Supplement to: Does interview language change how Hispanic adults report childhood household adversity?

Preprint version 0.1, 2 November 2026. All tables are generated from the files in `analysis/` and `exploratory/` of the repository; the source file is named under each table.

Contents

- S1. Full sample characteristics
- S2. Item prevalence with design-based CIs and prevalence ratios
- S3. Model fit
- S4. Multigroup CFA by interview language (S3 in the plan)
- S5. Anchor sensitivity and alignment (S4 in the plan)
- S6. Impact, mode and 11-item models (S5–S7 in the plan); deviations from the frozen plan
- S7. Exploratory analyses (not pre-specified)
- S8. Literature search summary

S1. Full sample characteristics

Source: `analysis/table1.md`. n = unweighted; % = weighted column percentage (BRFSS final weight ÷ 7; CHIS RAKEDW0 ÷ 5). Models use the native age groups (BRFSS _AGEG5YR 1–13 plus DK/refused; CHIS SRAGE_P1 14 groups); the groups below are collapsed for display.

BRFSS 2019–2025 (English n = 24,473; Spanish n = 10,945)

Characteristic	English n	English %	Spanish n	Spanish %
Age 18–24	4,175	21.7	747	8.5
25–34	5,186	22.9	2,126	19.0
35–44	4,536	18.9	2,687	23.6
45–54	3,731	13.9	2,154	18.3
55–64	3,202	12.3	1,565	15.0
65+	3,450	9.6	1,625	15.4
Don't know/refused	193	0.8	41	0.3
Sex Male	11,873	50.7	4,880	44.5
Female	12,600	49.3	6,065	55.5
Education Never attended/kindergarten	43	0.2	307	2.4
Grades 1–8	571	3.4	3,665	36.3
Grades 9–11	1,468	8.9	1,899	21.8
Grade 12/GED	7,425	29.9	2,856	23.4
College 1–3 years	7,218	33.1	948	8.6
College 4+ years	7,692	24.1	1,149	6.7
Refused	56	0.3	121	1.0
Household income <$15,000	1,644	6.8	2,115	22.2
$15,000–24,999	2,934	11.5	2,920	25.4
$25,000–34,999	2,664	10.7	1,694	14.2
$35,000–49,999	3,277	12.6	1,052	9.6
$50,000+	10,463	44.2	625	6.2

Characteristic	English n	English %	Spanish n	Spanish %
Missing/DK/refused	3,491	14.2	2,539	22.4
Survey year 2019	4,371	14.0	1,726	11.9
2020	7,607	36.2	2,995	40.6
2021	2,675	9.3	1,124	7.9
2022	2,581	10.9	1,214	9.7
2023	2,638	10.9	1,641	11.7
2024	2,634	11.3	883	9.1
2025	1,967	7.4	1,362	9.1
Census region Northeast	3,498	18.0	2,056	19.7
Midwest	5,038	12.0	2,368	6.8
South	8,638	42.8	4,441	43.1
West	7,299	27.2	2,080	30.5

CHIS 2021–2025 (English n = 24,418; Spanish n = 5,536)

Characteristic	English n	English %	Spanish n	Spanish %
Age 18–25	2,755	25.4	189	5.9
26–34	3,778	20.0	449	9.4
35–44	4,476	17.6	1,029	22.9
45–54	4,247	14.3	1,388	24.9
55–64	4,321	11.8	1,347	20.4
65+	4,841	10.9	1,134	16.5
Sex Male	9,789	49.7	2,190	47.1
Female	14,629	50.3	3,346	52.9
Education No formal education or grades 1–8	660	7.0	1,428	39.9
Grades 9–11	777	8.7	741	22.7
Grade 12/high school diploma	4,263	31.4	1,386	23.8
Some college	4,515	13.7	555	4.5
Vocational school	2,006	4.7	404	2.3
AA/AS degree	2,205	5.8	121	0.8
BA/BS degree or some graduate school	6,137	18.7	692	4.7
MA/MS degree	2,923	7.6	142	1.0
PhD or equivalent	932	2.4	67	0.4
Poverty level 0–99% FPL	3,659	18.5	1,848	35.0
100–199% FPL	4,305	20.4	1,833	32.7
200–299% FPL	3,509	15.4	845	15.2
300%+ FPL	12,945	45.7	1,010	17.1
Survey year 2021	4,921	20.2	1,105	20.3
2022	4,581	20.5	1,136	19.7
2023	4,934	20.8	1,064	19.3
2024	5,482	19.2	1,324	19.9
2025	4,500	19.3	907	20.8
Nativity Born in US	18,113	72.3	402	7.8

Characteristic	English n	English %	Spanish n	Spanish %
Born outside US	6,305	27.7	5,134	92.2
Percent of life in US 0–20%	230	1.2	606	10.3
21–40%	494	2.2	964	15.4
41–60%	1,094	5.8	1,959	36.6
61–80%	2,040	10.1	1,457	27.4
81–100% (incl. all US-born)	20,560	80.7	550	10.3
Interview mode Phone (interviewer)	3,104	13.3	1,844	34.1
Web (self-administered)	21,314	86.7	3,692	65.9
English proficiency Speaks only English	9,669	32.4	29	0.5
Very well/well	14,070	62.3	2,640	40.1
Not well/not at all	679	5.4	2,867	59.4

S2. Item prevalence with design-based CIs and prevalence ratios

Source: `analysis/table2.md`. Prevalence CIs: logit-transformed design-based intervals. PR = Spanish vs English prevalence ratio from a survey-weighted Poisson working model, Wald 95% CI. Adjusted: BRFSS age group, sex, education, income (missing as category), survey year, Census region; CHIS age group, sex, education, poverty level, survey year.

BRFSS 2019–2025 (Taylor linearization; strata = year × file × _STSTR, PSU = _PSU, lonely PSU = adjust)

Item	English % (95% CI)	Spanish % (95% CI)	Crude PR (95% CI)	Adjusted PR (95% CI)
Household mental illness	21.9 (20.8, 23.0)	4.0 (3.3, 4.9)	0.18 (0.15, 0.23)	0.24 (0.19, 0.31)
Household problem drinking	25.3 (24.2, 26.5)	16.8 (15.3, 18.4)	0.67 (0.60, 0.74)	0.68 (0.60, 0.77)
Household drug use	14.2 (13.3, 15.2)	3.1 (2.6, 3.7)	0.22 (0.18, 0.26)	0.21 (0.17, 0.26)
Household incarceration	14.0 (13.0, 15.0)	2.6 (2.1, 3.3)	0.19 (0.15, 0.24)	0.20 (0.16, 0.26)
Parental separation/divorce	39.0 (37.6, 40.3)	24.7 (23.1, 26.4)	0.63 (0.59, 0.68)	0.66 (0.60, 0.72)
Witnessed parental violence	23.1 (21.9, 24.3)	17.1 (15.6, 18.6)	0.74 (0.67, 0.82)	0.67 (0.59, 0.76)
Physical abuse	33.8 (32.5, 35.1)	30.4 (28.5, 32.3)	0.90 (0.83, 0.97)	0.88 (0.80, 0.97)
Verbal abuse	40.8 (39.4, 42.2)	18.1 (16.5, 19.8)	0.44 (0.40, 0.49)	0.51 (0.45, 0.57)
4+ ACEs (standard 8-category score)	25.3 (24.1, 26.5)	10.8 (9.5, 12.1)	0.43 (0.37, 0.48)	0.43 (0.37, 0.51)

CHIS 2021–2025 (jackknife replicate weights RAKEDW1–80, scale 1, MSE)

Item	English % (95% CI)	Spanish % (95% CI)	Crude PR (95% CI)	Adjusted PR (95% CI)	Adj. + nativity PR (95% CI)*
Household mental illness	23.8 (23.1, 24.5)	5.6 (4.9, 6.5)	0.24 (0.21, 0.27)	0.35 (0.30, 0.40)	0.45 (0.38, 0.53)
Household problem drinking	30.1 (29.2, 31.0)	18.4 (17.2, 19.5)	0.61 (0.57, 0.65)	0.60 (0.56, 0.65)	0.67 (0.62, 0.73)
Household drug use	14.5 (13.9, 15.1)	4.6 (3.9, 5.3)	0.31 (0.27, 0.37)	0.32 (0.27, 0.37)	0.49 (0.41, 0.58)
Household incarceration	13.1 (12.6, 13.7)	2.6 (2.1, 3.2)	0.20 (0.16, 0.25)	0.22 (0.17, 0.28)	0.43 (0.33, 0.57)
Parental separation/divorce	30.5 (29.6, 31.4)	21.0 (19.9, 22.1)	0.69 (0.65, 0.73)	0.70 (0.65, 0.75)	0.78 (0.71, 0.85)
Witnessed parental violence	27.6 (26.9, 28.3)	20.3 (19.1, 21.7)	0.74 (0.69, 0.79)	0.69 (0.63, 0.75)	0.71 (0.65, 0.78)
Physical abuse	28.5 (27.7, 29.4)	29.4 (28.1, 30.7)	1.03 (0.98, 1.08)	1.00 (0.94, 1.05)	0.98 (0.92, 1.05)
Verbal abuse	45.8 (44.9, 46.8)	27.3 (25.9, 28.7)	0.60 (0.56, 0.63)	0.67 (0.63, 0.70)	0.72 (0.67, 0.77)
4+ ACEs (standard 8-category score)	26.1 (25.4, 26.8)	13.6 (12.6, 14.8)	0.52 (0.48, 0.57)	0.56 (0.51, 0.62)	0.65 (0.58, 0.73)

*Not pre-specified for Table 2; added as context for H1b (adds born outside US and percent of life in US).

S3. Model fit

Source: `analysis/RESULTS.md` §3. WLSMV, theta parameterization, survey weights; scaled (mean- and variance-adjusted) indices. CFI/TLI are computed against a baseline conditional on the covariates. Fit was reported but was not a decision criterion.

Model	χ^2 (scaled)	df	CFI	TLI	RMSEA	SRMR
Primary MIMIC, BRFSS	1124.0	226	0.946	0.993	0.0106	0.050
Primary MIMIC, CHIS	1594.6	196	0.942	0.992	0.0154	0.050
S1 CHIS + nativity	1521.1	226	0.947	0.993	0.0138	0.050
S2 CHIS foreign-born	516.7	220	0.968	0.996	0.0109	0.065
S4a one factor, BRFSS	1535.5	262	0.924	0.992	0.0117	0.077
S4a one factor, CHIS	2625.8	227	0.901	0.988	0.0188	0.096
S6 CHIS mode interaction	1556.1	202	0.944	0.992	0.0150	0.050
S7 BRFSS 11 items, 3 factors	1281.2	318	0.975	0.996	0.0092	0.042

Primary and nativity models: all language parameters (source: `analysis/primary_results.md`, `analysis/secondary/effects_s1.csv`, `effects_s2.csv`; replicate SEs)

Model	Parameter	b (probit)	Replicate SE	95% CI	OR-equiv. (95% CI)	p	Δ endorsement, Spanish (pp, 95% CI)
BRFSS primary	Household dysfunction factor	−0.537	0.061	−0.656, −0.418		<0.001	
BRFSS primary	Physical/emotional abuse factor	−0.408	0.049	−0.504, −0.312		<0.001	
BRFSS primary	Household mental illness	−0.624	0.114	−0.848, −0.401	0.35 (0.24, 0.51)	<0.001	−6.0 (−7.9, −4.1)
BRFSS primary	Household drug use	−0.696	0.118	−0.927, −0.466	0.31 (0.21, 0.45)	<0.001	−3.1 (−4.2, −2.0)
BRFSS primary	Household incarceration	−0.655	0.088	−0.826, −0.483	0.33 (0.25, 0.44)	<0.001	−5.2 (−6.6, −3.8)
CHIS primary	Household dysfunction factor	−0.601	0.037	−0.673, −0.528		<0.001	
CHIS primary	Physical/emotional abuse factor	−0.270	0.025	−0.318, −0.222		<0.001	
CHIS primary	Household mental illness	−0.315	0.051	−0.416, −0.214	0.59 (0.49, 0.70)	<0.001	−4.4 (−5.8, −3.1)
CHIS primary	Household drug use	−0.266	0.082	−0.426, −0.106	0.64 (0.48, 0.83)	0.001	−1.5 (−2.4, −0.6)
CHIS primary	Household incarceration	−0.493	0.086	−0.662, −0.324	0.43 (0.32, 0.58)	<0.001	−2.9 (−3.8, −1.9)
S1 CHIS + nativity	Household dysfunction factor	−0.441	0.041			<0.001	
S1 CHIS + nativity	Physical/emotional abuse factor	−0.234	0.030			<0.001	
S1 CHIS + nativity	Household mental illness	−0.231	0.059		0.67 (0.55, 0.82)	<0.001	

Model	Parameter	b (probit)	Replicate SE	95% CI	OR-equiv. (95% CI)	p	Δ endorsement, Spanish (pp, 95% CI)
S1 CHIS + nativity	Household drug use	−0.066	0.091		0.89 (0.66, 1.21)	0.469	
S1 CHIS + nativity	Household incarceration	−0.128	0.106		0.81 (0.57, 1.15)	0.228	
S2 CHIS foreign-born (n = 11,439)	Household dysfunction factor	−0.375	0.056			<0.001	
S2 CHIS foreign-born	Physical/emotional abuse factor	−0.147	0.037			<0.001	
S2 CHIS foreign-born	Household mental illness	−0.217	0.072		0.69 (0.54, 0.88)	0.003	
S2 CHIS foreign-born	Household drug use	−0.005	0.112		0.99 (0.68, 1.44)	0.967	
S2 CHIS foreign-born	Household incarceration	0.002	0.129		1.00 (0.65, 1.54)	0.990	

Holm-adjusted p-values for the six primary target tests: all ≤ 0.001. Replicates converged: BRFSS 100/100, CHIS 80/80.

S4. Multigroup CFA by interview language (plan S3)

Source: `analysis/secondary/s3_fit.csv`, `s3_difftests.csv`, `s3_latent_means.csv`, `s3_score_tests.csv`. Binary items, theta parameterization, latent-response residual variances fixed at 1 in both groups. Scalar models: loadings and thresholds equal; Spanish latent means and variances free. Partial: as scalar but the three target items' loadings and thresholds free. Equality constraints were written explicitly because lavaan 0.7.2's `group.equal = "thresholds"` was not applied for these binary items (df checked: 38/50/44).

Survey	Model	χ^2 (scaled)	df	CFI	RMSEA	SRMR
BRFSS	configural	317.3	38	0.983	0.0204	0.052
BRFSS	scalar	586.1	50	0.967	0.0246	0.056
BRFSS	partial	529.9	44	0.970	0.0250	0.055
CHIS	configural	487.8	38	0.982	0.0281	0.051
CHIS	scalar	698.5	50	0.973	0.0294	0.055
CHIS	partial	633.5	44	0.976	0.0299	0.052

Survey	Comparison	Δχ^2 (scaled)	Δdf	p	ΔCFI	ΔRMSEA
BRFSS	configural vs full scalar	257.0	12	<0.001	−0.0160	0.0042
BRFSS	configural vs partial scalar (targets free)	190.4	6	<0.001	−0.0128	0.0046
BRFSS	partial scalar vs full scalar	79.5	6	<0.001	−0.0031	−0.0004
CHIS	configural vs full scalar	217.9	12	<0.001	−0.0081	0.0013
CHIS	configural vs partial scalar (targets free)	139.1	6	<0.001	−0.0057	0.0018
CHIS	partial scalar vs full scalar	93.6	6	<0.001	−0.0024	−0.0005

Latent mean of the Spanish-interview group (English = 0, SD 1): BRFSS scalar household dysfunction −0.605 (SE 0.052), abuse −0.453 (0.048); BRFSS partial −0.466 (0.060), −0.453 (0.048). CHIS scalar −0.674 (0.049), −0.449 (0.047); CHIS partial −0.578 (0.064), −0.449 (0.047).

Score tests on the full scalar model (eight largest per survey; one cross-group equality constraint per row):

Survey	Constraint	χ^2	scaled χ^2	p (scaled)
BRFSS	Verbal abuse: threshold	121.2	217.1	<0.001
BRFSS	Household problem drinking: threshold	112.7	129.5	<0.001
BRFSS	Physical abuse: threshold	59.5	102.1	<0.001
BRFSS	Household mental illness: threshold	54.7	61.2	<0.001
BRFSS	Household problem drinking: loading	31.6	22.7	<0.001
BRFSS	Verbal abuse: loading	25.0	38.2	<0.001
BRFSS	Witnessed parental violence: threshold	13.4	22.9	<0.001
BRFSS	Household incarceration: threshold	11.1	12.1	<0.001
CHIS	Physical abuse: threshold	100.2	207.9	<0.001
CHIS	Household mental illness: threshold	83.1	90.8	<0.001
CHIS	Verbal abuse: threshold	82.5	147.3	<0.001
CHIS	Household problem drinking: threshold	59.7	75.5	<0.001
CHIS	Physical abuse: loading	38.8	56.4	<0.001
CHIS	Household incarceration: loading	22.3	14.2	<0.001
CHIS	Household problem drinking: loading	17.5	13.7	<0.001
CHIS	Verbal abuse: loading	12.0	15.0	<0.001

S5. Anchor sensitivity and alignment (plan S4)

Source: `analysis/secondary/s4_effects.csv`, `s4b_purification_history.csv`, `s4d_alignment_*.csv`. lavaan weight-robust SEs.

Survey	Analysis	Item	b	SE	p	OR-equiv. (95% CI)
BRFSS	S4c one-at-a-time	Household problem drinking	0.606	0.128	<0.001	2.80 (1.83, 4.29)
BRFSS	S4c one-at-a-time	Parental separation/divorce	−0.255	0.050	<0.001	0.65 (0.55, 0.77)
BRFSS	S4c one-at-a-time	Witnessed parental violence	0.085	0.077	0.269	1.16 (0.89, 1.49)
BRFSS	S4c one-at-a-time	Physical abuse	0.419	0.057	<0.001	2.04 (1.69, 2.46)
BRFSS	S4c one-at-a-time	Verbal abuse	−0.691	0.078	<0.001	0.31 (0.24, 0.40)
BRFSS	S4a one factor	Household mental illness	−0.677	0.086	<0.001	0.32 (0.24, 0.42)
BRFSS	S4a one factor	Household drug use	−0.745	0.094	<0.001	0.28 (0.21, 0.39)
BRFSS	S4a one factor	Household incarceration	−0.708	0.087	<0.001	0.30 (0.22, 0.40)
BRFSS	S4b purified (freed: verbal, drinking, witnessed violence)	Household mental illness	−0.223	0.139	0.110	0.68 (0.43, 1.09)
BRFSS	S4b purified	Household drug use	−0.137	0.169	0.418	0.79 (0.45, 1.39)
BRFSS	S4b purified	Household incarceration	−0.219	0.138	0.113	0.69 (0.43, 1.09)
BRFSS	S4b purified	Verbal abuse	−0.872	0.089	<0.001	0.23 (0.17, 0.30)
BRFSS	S4b purified	Household problem drinking	0.606	0.128	<0.001	2.80 (1.83, 4.30)
BRFSS	S4b purified	Witnessed parental violence	−0.289	0.085	<0.001	0.61 (0.46, 0.81)
CHIS	S4c one-at-a-time	Household problem drinking	0.247	0.125	0.048	1.52 (1.00, 2.31)

Survey	Analysis	Item	b	SE	p	OR-equiv. (95% CI)
CHIS	S4c one-at-a-time	Parental separation/divorce	−0.092	0.045	0.042	0.86 (0.74, 0.99)
CHIS	S4c one-at-a-time	Witnessed parental violence	−0.186	0.057	0.001	0.73 (0.60, 0.88)
CHIS	S4c one-at-a-time	Physical abuse	0.687	0.065	<0.001	3.21 (2.59, 3.99)
CHIS	S4c one-at-a-time	Verbal abuse	−0.471	0.060	<0.001	0.45 (0.37, 0.55)
CHIS	S4a one factor	Household mental illness	−0.517	0.058	<0.001	0.42 (0.34, 0.50)
CHIS	S4a one factor	Household drug use	−0.547	0.082	<0.001	0.39 (0.30, 0.52)
CHIS	S4a one factor	Household incarceration	−0.730	0.092	<0.001	0.29 (0.21, 0.39)
CHIS	S4b purified (freed: physical abuse)	Household mental illness	−0.315	0.069	<0.001	0.59 (0.47, 0.74)
CHIS	S4b purified	Household drug use	−0.266	0.102	0.009	0.64 (0.45, 0.89)
CHIS	S4b purified	Household incarceration	−0.493	0.109	<0.001	0.43 (0.30, 0.62)
CHIS	S4b purified	Physical abuse	0.687	0.065	<0.001	3.21 (2.59, 3.99)
CHIS + nativity	S4c one-at-a-time	Household problem drinking	0.133	0.150	0.374	1.25 (0.76, 2.06)
CHIS + nativity	S4c one-at-a-time	Parental separation/divorce	−0.048	0.054	0.370	0.92 (0.77, 1.10)
CHIS + nativity	S4c one-at-a-time	Witnessed parental violence	−0.191	0.070	0.006	0.72 (0.57, 0.91)
CHIS + nativity	S4c one-at-a-time	Physical abuse	0.557	0.076	<0.001	2.58 (2.00, 3.32)
CHIS + nativity	S4c one-at-a-time	Verbal abuse	−0.331	0.071	<0.001	0.57 (0.45, 0.72)

Purification history (flag: $p < .01$ and $|b| \geq 0.24$; the largest flagged $|b|$ freed each iteration, at least one anchor kept per factor): BRFSS iteration 1 flagged drinking, divorce, physical abuse, verbal abuse → freed verbal; iteration 2 flagged drinking, divorce, witnessed violence → freed drinking; iteration 3 flagged witnessed violence → freed; iteration 4 none. CHIS iteration 1 flagged physical abuse, verbal abuse → freed physical abuse; iteration 2 none.

Alignment (anchor-free; Spanish − English aligned intercept on the probit scale; `sirt::invariance.alignment` on the configural two-group estimates, no covariates):

Survey	Factor	Item	Spanish − English
BRFSS	Household dysfunction	Household mental illness	−0.47
BRFSS	Household dysfunction	Household problem drinking	0.47
BRFSS	Household dysfunction	Household drug use	0.01
BRFSS	Household dysfunction	Household incarceration	−0.21
BRFSS	Household dysfunction	Parental separation/divorce	−0.16
BRFSS	Abuse	Witnessed parental violence	−0.01
BRFSS	Abuse	Physical abuse	0.05
BRFSS	Abuse	Verbal abuse	−1.02
CHIS	Household dysfunction	Household mental illness	−0.65
CHIS	Household dysfunction	Household problem drinking	0.00
CHIS	Household dysfunction	Household drug use	0.02
CHIS	Household dysfunction	Household incarceration	−0.18

Survey	Factor	Item	Spanish − English
CHIS	Household dysfunction	Parental separation/divorce	−0.11
CHIS	Abuse	Witnessed parental violence	0.00
CHIS	Abuse	Physical abuse	0.52
CHIS	Abuse	Verbal abuse	−0.43

Aligned Spanish latent means (SD): BRFSS household dysfunction −0.578 (0.736), abuse −0.225 (1.078); CHIS household dysfunction −0.498 (0.905), abuse −0.409 (1.180). With only two groups alignment is weakly informative and is reported as a descriptive check.

S6. Impact, mode and 11-item models; deviations from the plan

S5 impact (source: `analysis/secondary/s5_impact_*.csv`; 8 harmonized items; 50 simulation draws per respondent with common random numbers; replicate 95% CIs)

Survey	Quantity	Observed	Model-implied (as estimated)	Target DIF removed	Change in gap	% of model-implied gap due to target DIF
BRFSS	Gap in mean 8-item count (Sp − En)	−0.95 (−1.03, −0.87)	−0.94 (−1.03, −0.86)	−0.80 (−0.90, −0.70)	0.14 (0.11, 0.18)	15.3 (11.1, 19.6)
BRFSS	Gap in % with ≥ 4 of 8 items (pp)	−14.8 (−16.5, −13.2)	−15.0 (−16.2, −13.8)	−12.2 (−13.7, −10.8)	2.7 (2.1, 3.4)	18.3 (13.4, 23.2)
CHIS	Gap in mean 8-item count (Sp − En)	−0.85 (−0.91, −0.79)	−0.84 (−0.90, −0.78)	−0.75 (−0.82, −0.68)	0.09 (0.07, 0.11)	10.3 (7.6, 13.1)
CHIS	Gap in % with ≥ 4 of 8 items (pp)	−12.7 (−14.0, −11.5)	−13.7 (−14.6, −12.7)	−12.1 (−13.1, −11.0)	1.6 (1.2, 2.0)	11.8 (8.6, 15.0)

Spanish mean count / % ≥ 4: BRFSS observed 1.17 / 9.0%, model 1.19 / 8.6%, DIF removed 1.33 / 11.3% (English observed 2.12 / 23.8%). CHIS observed 1.29 / 11.1%, model 1.31 / 10.1%, DIF removed 1.39 / 11.7% (English observed 2.14 / 23.8%).

S6 mode (CHIS) (source: `analysis/secondary/s6_mode.csv`; Spanish × web interaction on the target direct effects; robust SEs)

Item	Term	b	SE	p	OR-equiv. (95% CI)
Household mental illness	Spanish direct effect, phone	−0.252	0.128	0.049	0.65 (0.43, 1.00)
Household drug use	Spanish direct effect, phone	−0.134	0.169	0.428	0.80 (0.45, 1.40)
Household incarceration	Spanish direct effect, phone	−0.144	0.173	0.404	0.78 (0.44, 1.39)
Household mental illness	Spanish direct effect, web	−0.316	0.077	<0.001	0.58 (0.45, 0.76)
Household drug use	Spanish direct effect, web	−0.306	0.117	0.009	0.59 (0.40, 0.88)
Household incarceration	Spanish direct effect, web	−0.687	0.134	<0.001	0.31 (0.20, 0.49)
Household mental illness	Spanish × web interaction	−0.064	0.141	0.649	0.90 (0.56, 1.44)
Household drug use	Spanish × web interaction	−0.172	0.190	0.365	0.75 (0.40, 1.40)
Household incarceration	Spanish × web interaction	−0.543	0.209	0.009	0.40 (0.20, 0.80)
Household mental illness	Web direct effect (English)	0.129	0.065	0.048	1.24 (1.00, 1.55)
Household drug use	Web direct effect (English)	0.039	0.087	0.654	1.07 (0.80, 1.43)
Household incarceration	Web direct effect (English)	−0.079	0.082	0.338	0.88 (0.67, 1.15)

S7 BRFSS 11 items, three-factor MIMIC (source: `analysis/secondary/s7_effects.csv`; robust SEs)

Parameter	b	SE	p	OR-equiv. (95% CI)	Δ endorsement (pp)
Physical/emotional abuse factor	−0.408	0.047	<0.001		
Household dysfunction factor	−0.534	0.056	<0.001		
Sexual abuse factor	−0.406	0.049	<0.001		
Household drug use	−0.695	0.108	<0.001	0.31 (0.21, 0.44)	−3.1
Household incarceration	−0.655	0.092	<0.001	0.33 (0.24, 0.45)	−5.4
Household mental illness	−0.630	0.102	<0.001	0.34 (0.24, 0.48)	−5.9

Deviations from the frozen plan (full text with reasons: `analysis/DEVIATIONS.md`)

1. BRFSS bootstrap: 749 single-PSU strata (of 2,837) were merged with the nearest multi-PSU stratum in the same year × file × state (or year × file) because Rao–Wu subbootstrap weights are undefined for 1-PSU strata; 2,088 bootstrap strata remained. Taylor-linearization analyses (Table 2) use the planned strata with lonely-PSU adjustment. Check: bootstrap SE of the weighted Spanish share 0.0054 vs 0.0058 by Taylor; of mental-illness prevalence 0.0044 vs 0.0041.
2. Integer PSU and stratum IDs were passed to `survey::as.svrepdesign` for speed; the algorithm is unchanged.
3. Clarifications fixed before fitting: BRFSS age DK/refused and education refused kept as categories; replicate variance with MSE = TRUE; two-sided Wald p; the partial-support rule read as the same items in both surveys (the alternative reading gives the same verdict); S3 binary-item identification; S4b one-item-per-iteration purification with at least one anchor per factor; S6 includes web main effects; S5 uses the 8 harmonized items; one labelled, non-pre-specified CHIS + nativity column added to Table 2.
4. Interpretive note (not a deviation): the plan assumed anchor bias could only be conservative; S4c shows problem drinking is over-endorsed in Spanish in BRFSS, so the primary BRFSS target estimates are not guaranteed to be conservative.

S7. Exploratory analyses (not pre-specified; run 1 November 2026)

These analyses preceded the frozen plan and generated H1. They use approximate design-based variance (cluster-robust sandwich estimators; state × year or year × stratum × PSU clusters; no replicate weights) and were not corrected for multiplicity across analyses. Full results, scripts and caveats: `exploratory/*/RESULTS.md`, `scripts/exploratory/`.

S7.1 BRFSS: logistic-regression DIF, two-parameter logistic anchor sensitivity, and the reconciliation of estimands

Source: `exploratory/dif/RESULTS.md`. Sample as in the main text (10,945 / 24,473), weights normalised to mean 1 within year.

Logistic-regression DIF, matching on the rest score of the other 10 items (no covariates). M1: item ~ S; M2: + Spanish; M3: + Spanish × S. Flag: BH-FDR q < .05 across 22 tests and ΔR^2 (M1 → M2) ≥ .02 or ≥ 10% change in the rest-score coefficient.

Item	Sp %	En %	Crude OR	Uniform-DIF OR [95% CI]	pp diff at matched score	ΔR^2 uniform	ΔR^2 non-uniform	Flag
Household mental illness	4.2	22.2	0.15	**0.21 [0.15, 0.29]**	−10.6	0.052	0.000	yes
Household problem drinker	16.8	25.5	0.59	1.06 [0.99, 1.12]	+0.6	0.000	0.001	–
Household drug use	3.2	14.5	0.19	**0.33 [0.24, 0.44]**	−4.7	0.021	0.000	yes

Item	Sp %	En %	Crude OR	Uniform-DIF OR [95% CI]	pp diff at matched score	ΔR^2 uniform	ΔR^2 non-uniform	Flag
Household incarceration	2.7	14.0	0.17	**0.27 [0.19, 0.37]**	−5.8	0.031	0.000	yes
Parents separated/divorced	25.4	39.2	0.53	0.67 [0.54, 0.82]	−8.1	0.009	0.001	–
Witnessed parental violence	17.1	23.1	0.69	1.42 [1.25, 1.61]	+3.7	0.004	0.002	–
Physical abuse	30.2	34.1	0.84	1.50 [1.23, 1.83]	+6.9	0.007	0.005	–
Verbal abuse	17.7	40.9	0.31	**0.42 [0.31, 0.55]**	−12.3	0.024	0.003	yes
Sexual: touched	6.8	13.9	0.45	1.06 [0.95, 1.19]	+0.3	0.000	0.008	–
Sexual: made to touch	6.0	10.7	0.53	1.34 [1.16, 1.53]	+1.2	0.002	0.005	–
Sexual: forced sex	4.4	6.8	0.64	1.98 [1.56, 2.53]	+1.9	0.009	0.005	–

With covariates (age, education, sex, income, year, state) the four ORs were 0.33, 0.34, 0.30 and 0.54; unweighted and purified (7-item matching) analyses flagged the same four items; recoding "parents not married" changed no OR by more than 0.02. Non-uniform DIF was negligible throughout (all $\Delta R^2 < .01$).

Two-group 2PL under alternative anchor sets (uniform DIF OR at equal latent level; anchors constrained equal across groups; Spanish latent mean and SD free; point estimates only):

Item	A1 purified 7	A2 drinker only	A3 drink + divorce + sexual	A4 household 5	X1 physical + witnessed (extreme)	X2 verbal + mental illness (extreme)
Household mental illness	0.17	0.20	0.19	*anchor*	0.14	*anchor*
Household problem drinker	*anchor*	*anchor*	*anchor*	*anchor*	0.71	2.67
Household drug use	0.24	0.28	0.27	*anchor*	0.19	0.91
Household incarceration	0.22	0.24	0.24	*anchor*	0.18	0.61
Parents separated/divorced	*anchor*	0.67	*anchor*	*anchor*	0.58	1.02
Witnessed parental violence	*anchor*	1.30	1.32	3.63	*anchor*	3.97
Physical abuse	*anchor*	1.58	1.65	3.78	*anchor*	4.06
Verbal abuse	0.33	0.39	0.40	1.23	0.25	*anchor*
Sexual: touched	*anchor*	0.95	*anchor*	8.27	0.45	9.22
Sexual: made to touch	*anchor*	1.31	*anchor*	11.87	0.61	13.08
Sexual: forced sex	*anchor*	1.91	*anchor*	16.82	0.90	18.64
Spanish latent mean (English SD units)	−0.38	−0.46	−0.52	−1.00	−0.23	−1.06

The items fall into two clusters with different language gaps. Under anchors drawn from the small-gap cluster (A1–A3), the four large-gap items have ORs of 0.17–0.40 and the latent deficit is 0.4–0.5 SD; under household anchors (A4), the abuse and sexual items appear over-endorsed by ORs of 3.6–16.8 and the deficit is 1.0 SD. Internal data cannot choose between these. DIF-corrected Spanish 4+ prevalence ranges from 5.4% to 19.3% across anchor sets (observed 10.7%; English 25.5%); under A1–A3 it is 13.8–15.8%.

Reconciling the odds ratios across analyses. For household mental illness in BRFSS the repository reports a crude OR of 0.17, a rest-score logistic DIF OR of 0.21 (0.33 with covariates), 2PL ORs of 0.17–0.20, a MIMIC primary OR-equivalent of 0.35, and a MIMIC purified OR-equivalent of 0.68. These differ because they are different estimands: the logistic and 2PL analyses condition on a single dimension defined by all other items (including the abuse and sexual items, which anchor the latent deficit at 0.4–0.5 SD), whereas the MIMIC conditions on a separate household-dysfunction factor whose language effect is identified by drinking and divorce only, adjusts for covariates, and reports $\exp(1.7\beta)$ from a probit model. The purified MIMIC is relative to

divorce alone. The ordering is the same in every analysis: mental illness, incarceration, drug use and verbal abuse are the most under-endorsed items relative to the rest.

Spanish wording period. The 2019 and 2020 CDC Spanish versions differed on the witnessed-violence item, and the 2021–2025 Spanish text is unpublished. The witnessed-violence DIF fell from 2019 to 2020 (OR ratio 0.64–0.76, p = .005–.08) as the wording change would predict, but items with unchanged wording moved as much (verbal abuse ×1.4–1.6; forced sex ×0.55–0.66). The four flagged items showed the same direction in all three periods. No wording effect is identifiable.

S7.2 BRFSS: respondents' own mental-health reports versus the household item

Source: `exploratory/brfss_mh/RESULTS.md`. Sample A: Hispanic ACE-module respondents 2019–2025 (11,995 Spanish / 27,468 English; all valid responses per item). Adjusted for age, sex, education, income, state and year; clusters = year × stratum × PSU; weights normalised within year.

Measure	wt% Spanish / English	Crude OR (95% CI)	Adjusted OR (95% CI)
ACE: lived with depressed/mentally ill/suicidal	4.4 / 22.8	0.16 (0.13–0.19)	0.20 (0.15–0.25)
ACE: lived with someone incarcerated	2.8 / 14.5	0.17 (0.14–0.21)	0.16 (0.12–0.21)
ACE: lived with drug user	3.4 / 14.8	0.20 (0.17–0.25)	0.20 (0.16–0.24)
ACE: sworn at/insulted	18.4 / 41.8	0.32 (0.28–0.35)	0.37 (0.32–0.42)
ACE: physically hurt by parent/adult	31.3 / 35.0	0.84 (0.77–0.93)	0.80 (0.70–0.91)
Own depression diagnosis	9.3 / 19.3	0.43 (0.37–0.49)	0.33 (0.28–0.40)
Frequent mental distress (≥ 14 poor mental-health days)	10.5 / 16.1	0.61 (0.53–0.70)	0.51 (0.43–0.62)
PHQ-8 ≥ 10 (2010–11 module, non-ACE states)	9.4 / 11.8	0.78 (0.52–1.16)	0.32 (0.21–0.49)
Fair/poor general health	35.0 / 16.6	2.70 (2.44–2.99)	1.18 (1.02–1.36)

Ratio of adjusted ORs, household mental illness vs comparator, same respondents, stacked model: vs own depression diagnosis 0.59 (0.45–0.78); vs frequent mental distress 0.37 (0.29–0.48); vs physical abuse 0.24 (0.19–0.32); vs incarceration 1.22 (0.89–1.67). Adding own depression diagnosis and frequent mental distress to the household-item model moved its aOR from 0.19 to 0.25. The same item cluster replicated in BRFSS 2009–2012 (mental illness 0.24, drugs 0.23, incarceration 0.18), where Spanish respondents reported *more* physical abuse (aOR 1.44). Puerto Rico 2024 (Spanish interviews, n ≈ 4,150, descriptive): household mental illness 12.6%, own depression 18.1%, versus mainland Spanish 4.4% and 9.3%; Spanish administration per se does not suppress endorsement to mainland-Spanish levels.

Differential prediction (weighted logistic regression of the outcome on predictor × Spanish plus covariates; BRFSS complete-case sample):

Outcome	Predictor	OR English (95% CI)	OR Spanish (95% CI)	Interaction p
Own depression diagnosis	ACE score (0–8)	1.40 (1.36–1.44)	1.41 (1.30–1.52)	0.92
Own depression diagnosis	Household mental-illness item	4.63 (3.99–5.36)	5.93 (3.70–9.51)	0.33
Frequent mental distress	ACE score (0–8)	1.31 (1.27–1.35)	1.35 (1.24–1.46)	0.51
Frequent mental distress	Household mental-illness item	3.20 (2.76–3.70)	4.14 (2.36–7.28)	0.38

S7.3 CHIS: exploratory odds ratios, mode and differential prediction

Source: `exploratory/chis/RESULTS.md`. Jackknife replicate SEs. Covariates: age, sex, education, poverty level, year; "+ nativity" adds US-born and percent of life in the US. The DIF column conditions on the sum of the other seven non-sexual items.

Item	Sp %	En %	aOR	aOR + nativity (95% CI)	aOR, foreign-born only	Logistic DIF OR (rest score + nativity)
Household mental illness	5.6	23.8	0.30	0.41 (0.34–0.49)	0.44 (0.35–0.55)	0.50 (0.40–0.62)
Household drug use	4.6	14.5	0.28	0.45 (0.38–0.55)	0.56 (0.43–0.73)	0.63 (0.50–0.79)
Household incarceration	2.6	13.1	0.19	0.40 (0.30–0.53)	0.54 (0.35–0.82)	0.57 (0.41–0.80)
Household problem drinker	18.4	30.1	0.51	0.59 (0.54–0.66)	0.64 (0.56–0.73)	0.74 (0.65–0.83)
Parents separated/divorced	21.0	30.5	0.61	0.71 (0.63–0.80)	0.78 (0.67–0.91)	0.80 (0.71–0.91)
Witnessed parental violence	20.4	27.6	0.61	0.63 (0.56–0.72)	0.69 (0.60–0.80)	0.84 (0.73–0.96)
Physical abuse	29.4	28.5	1.00	0.97 (0.88–1.07)	1.09 (0.96–1.23)	1.51 (1.33–1.71)
Verbal abuse	27.3	45.8	0.53	0.60 (0.54–0.66)	0.70 (0.62–0.80)	0.74 (0.66–0.84)
4+ ACEs	13.6	26.1	0.48	0.58 (0.51–0.67)	0.72 (0.61–0.86)	—
Serious psychological distress (K6, past month)	4.9	11.2	0.48	0.58 (0.48–0.71)	0.65 (0.52–0.81)	—
Fair/poor general health	35.9	17.8	1.38	1.73 (1.53–1.95)	1.81 (1.56–2.10)	—

Spanish-vs-English ORs within mode (adjusted incl. nativity): household mental illness 0.31 phone / 0.44 web (interaction p = .13); drug use 0.29 / 0.48 (.44); incarceration 0.43 / 0.32 (.08); physical abuse 1.08 / 0.95 (.56). Differential prediction of serious psychological distress: OR per ACE 1.37 (English) vs 1.45 (Spanish), p = .29; household mental-illness item 3.29 vs 3.10, p = .75.

S7.4 NSCH 2022–2024: Hispanic children in Spanish- versus English-primary-language households

Source: `exploratory/nsch_chis/RESULTS.md`. Census NSCH topical public-use files (current edited releases). Hispanic children (SC_HISPANIC_R = 1) in English- or Spanish-primary-language households (HHLANGUAGE; other/missing excluded, n = 756). Pooled 2022–2024: 16,676 English and 6,919 Spanish households (weighted Spanish share 40.4%). Weighted logistic regression (FWC) with sandwich SEs clustered on state (51 clusters; strata ignored); poverty level multiply imputed (6 implicates, Rubin's rules). M3 (primary): child age group, sex, year, highest adult education, poverty level and household generation. Sensitivity: first- and second-generation households only. Item contrasts: ratio of the M3 ORs for two items with a joint cluster sandwich.

Language and generation are heavily confounded: 0.4% of English-language vs 14.2% of Spanish-language households were first-generation, 27.9% vs 68.5% second-generation; 78.8% of responding adults in Spanish-language households were born outside the US; 38.9% had less than a high-school education (vs 6.0%).

Measure	English HH %	Spanish HH %	Crude OR	aOR M3 (95% CI)	aOR, 1st/2nd-gen HH only
ACE8: lived with anyone mentally ill, suicidal, severely depressed	9.6	2.9	0.28 (0.21–0.36)	**0.32 (0.21–0.49)**	0.34 (0.20–0.57)
ACE9: lived with anyone with alcohol/drug problem	9.6	4.5	0.44 (0.33–0.58)	0.47 (0.32–0.69)	0.59 (0.40–0.87)
ACE6: saw/heard adults hit each other at home	6.3	2.7	0.40 (0.33–0.49)	0.38 (0.31–0.45)	0.46 (0.35–0.62)
ACE5: parent/guardian served time in jail	7.5	2.5	0.32 (0.26–0.38)	0.33 (0.24–0.43)	0.36 (0.24–0.54)
ACE3: parent/guardian divorced/separated	25.8	19.7	0.71 (0.58–0.87)	0.73 (0.61–0.88)	0.81 (0.67–0.98)
ACE4: parent/guardian died	3.2	2.4	0.75 (0.54–1.04)	0.58 (0.43–0.78)	0.91 (0.58–1.44)
ACE7: victim/witness of neighborhood violence	4.8	4.3	0.88 (0.68–1.14)	0.57 (0.36–0.88)	0.93 (0.52–1.66)

Measure	English HH %	Spanish HH %	Crude OR	aOR M3 (95% CI)	aOR, 1st/2nd-gen HH only
ACE10: treated unfairly because of race/ethnicity	5.4	3.8	0.70 (0.57–0.85)	0.77 (0.62–0.96)	0.97 (0.72–1.29)
ACE1: hard to cover basics somewhat/very often	17.0	19.3	1.16 (1.07–1.26)	0.85 (0.73–0.99)	1.04 (0.88–1.23)
Responding adult's own mental health fair/poor	9.1	4.1	0.43 (0.36–0.50)	**0.39 (0.32–0.47)**	0.46 (0.36–0.58)
Comparator: responding adult's physical health fair/poor	9.1	9.8	1.08 (0.91–1.29)	0.69 (0.57–0.83)	0.79 (0.68–0.93)
Child depression, ever (age 3–17)	5.8	4.5	0.77 (0.52–1.13)	1.03 (0.78–1.36)	1.36 (0.93–1.97)
Child anxiety, current (age 3–17)	11.0	5.6	0.48 (0.39–0.59)	0.59 (0.48–0.73)	0.73 (0.60–0.88)
Comparator: ADHD, ever (age 3–17)	11.3	5.9	0.49 (0.35–0.68)	0.64 (0.42–0.98)	0.85 (0.54–1.32)
Comparator: asthma, ever	11.0	9.2	0.82 (0.76–0.89)	0.99 (0.84–1.17)	1.01 (0.86–1.18)

ACE8 by year (English vs Spanish households, weighted %): 2022 10.1 vs 3.4; 2023 10.0 vs 2.9; 2024 8.5 vs 2.4.

Item contrasts (M3, ratio of ORs): ACE8 vs ACE9 (alcohol/drugs) 0.68 (0.52–0.89, p = .005); ACE8 vs ACE5 (jail) 0.98 (0.62–1.55); ACE8 vs ACE6 (domestic violence) 0.85 (0.51–1.42); ACE8 vs ACE3 (divorce) 0.44 (0.33–0.58); ACE8 vs mean of all nine other ACE items 0.53 (0.36–0.78, p = .001); parent mental health fair/poor vs parent physical health fair/poor 0.56 (0.47–0.67, p < .001); child current anxiety vs asthma 0.60 (0.50–0.72); child anxiety vs ADHD 0.92 (0.62–1.38).

Item nonresponse: Spanish-language households skipped every ACE item about 2 percentage points more often than English-language households (ACE8 7.0% vs 5.2%; other ACE items 7.0–7.6% vs 5.0–5.6%; parent mental health 6.6% vs 4.5%), with no mental-health-specific skipping.

Caveats: household language is a proxy for the language the questionnaire was completed in; NSCH ACEs are parent reports about the child's lifetime and the household item can include the respondent; NSCH has no abuse items; generation is a coarse acculturation adjustment and nearly collinear with language.

S8. Literature search summary

Full logs with query strings and hit counts: `notes/01_novelty_verification_2026-11-01.md` and `notes/05_novelty_H1_2026-11-02.md`. Searches (1–2 November 2026): PubMed (E-utilities, 18 queries), Europe PMC full text including preprints (13 queries), OpenAlex (works, dissertations, preprints), Crossref posted content, Harvard DASH, and general web searches including Spanish- and Portuguese-language terms. SciELO, LILACS/BVS and Redalyc blocked programmatic access and were covered only indirectly; ProQuest was not searched directly.

Novelty component	Verdict	Closest prior work
Item-level language DIF of ACE items in BRFSS	Not found	Alhalel 2018 (marginal prevalence by questionnaire language); Mei 2022, Ford 2014 (invariance, language untested); Umoga 2026 (race/ethnicity, embargoed); Zhang & Lippert 2026 (2PL scoring; full text unread)
Language DIF of ACE items in CHIS	Not found; no psychometric study of the CHIS ACE items	Kim 2016 (K6 DIF by language in CHIS)

Novelty component	Verdict	Closest prior work
Negative DIF concentrated in household mental illness, drug use and incarceration	Not found as a conditional finding or as a named mechanism	Same-direction prevalence precursors (Alhalel 2018; Grest 2021); contrary: Jensen 2024 (no raised Hispanic threshold for household mental illness in NSCH), LaBrenz 2019 (withdrawn SSWR abstract; no language difference on household items, unconditional tests), HCHS/SOL high endorsement (Llabre 2017)
Replication across independent Spanish translations	Not found	MacFadyen 2023 (qualitative Spanish cognitive testing of a BRFSS-derived PRAMS item set; no Spanish-specific problems)
Language DIF persisting after nativity adjustment	Not found for ACE items	Cooper & Nickodem 2021 (all-foreign-born; nativity constant); Kobayashi 2024 (language and nativity as covariates only)

Residual risks to the novelty claim: the embargoed Umoga 2026 dissertation (full text unavailable until April 2027) and Zhang & Lippert 2026 (paywalled; abstract reports IRT scoring without group-specific parameters).